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A new key pole for the East European Craton at 1452 Ma: Palaeomagnetic and geochronological constraints from mafic rocks in the Lake Ladoga region (Russian Karelia)

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ABSTRACT

Palaeomagnetic and geochronological studies on mafic rocks in the Lake Ladoga region in South Russian Karelia provide a new, reliably dated Mesoproterozoic key paleopole for the East European Craton (Baltica). U–Pb dating on baddeleyite gives a crystallisation age of 1452 ± 12 Ma for one of the studied dolerite dykes. A mean palaeomagnetic pole for the Mesoproterozoic dolerite dykes, Valaam sill and Salmi basalts yields a paleopole at 15.2°N, 177.1°E, A_{95} = 5.5°. Positive baked contact test for the dolerite dykes and positive reversal test for the Salmi basalts and for the dykes confirm the primary nature of the magnetisation. Comparison of this Baltica palaeopole with coeval paleomagnetic data for Laurentia and Siberia provides a revised palaeoposition of these cratons. The results verify that the East European Craton, Laurentia and Siberia were part of the supercontinent Columbia from the Late Palaeoproterozoic to the Middle Neoproterozoic.

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1. Introduction

The palaeogeographic position of the East European Craton (Baltica) and its relationships with other continental blocks in the late Palaeoproterozoic and the earliest Mesoproterozoic have been under active discussions. In the Mid-Proterozoic NENA (=North Europe–North Atlantic) model of Gower et al. (1990), the northern margin (the present coordinates) of Baltica was connected to southern Greenland and eastern Laurentia based on geological correlations. Similar reconstructions were proposed by Park (1992, 1994), Gorbatschev and Bogdanova (1993), and by Hoffman (1997) for the 1.8 Ga Nuna megacontinent. Quite different Laurentia-Baltica connections were proposed by Rogers and Santosh (2002) and Zhao et al. (2004) within the Palaeoproterozoic–Mesoproterozoic supercontinent Columbia.

However, while there is a wide agreement that the Columbia supercontinent was formed at ca. 1.8 Ga, little is known about its later evolution prior to ca. 1.3 Ga when the supercontinent Rodinia began to assemble (Li et al., 2008). In some studies, it is proposed that Columbia began to break up between 1.6 and 1.5 Ga (e.g. Rogers and Santosh, 2002; Condie, 2002) and persisted until 1.3–1.2 Ga (Zhao et al., 2004), but this view has not been tested strictly with palaeomagnetic data. Condie (2002) suggested that large megacontinental assemblages kept their Palaeoproterozoic relationships to the time when Rodinia started to form. This seems to be true for some intercontinental connections such as Laurentia-Siberia (Pisarevsky et al., 2008; Wingate et al., 2009) or Baltica-Laurentia-Siberia-North China (Wu et al., 2005), which preserved their tectonic continuity in the Paleoproterozoic and features that show no 1.3–1.0 Ga active continental margins. However, the interrelationships between most continental blocks in the earliest Mesoproterozoic are controversial and still not assessed palaeomagnetically.

At present, reconstructions of Precambrian palaeogeography are based on 'key' palaeomagnetic poles, which pass basic reliability criteria and are precisely and accurately dated (Buchan et al., 2000). So far there is not yet a 'key' pole between the 1.78 and 1.25 Ga key poles for Baltica, and this hampers the appraisal of plate configuration and relative motions of cratons during the ca. 500 Ma-long

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interval. Correspondingly, the history of Columbia is still undefined for that period.

The published palaeomagnetic reconstructions involving Baltica differ depending on whether normal or reversed polarities are used for matching. Pesonen et al. (2003) proposed that Laurentia and Baltica were connected during the entire period between 1.83 and 1.27 Ga, and together with Siberia and perhaps some other continents like Australia, formed the core of the Columbia supercontinent. In the Columbia and Rodinia reconstructions of Meert (2002) Baltica has different positions in relation to Laurentia. For 1.5 Ga, Meert preferred reversed polarity for Baltica and some rotation for all involved continents such as Baltica, Laurentia, Siberia and Australia. He located Baltica in near-equatorial latitudes.

In a recent palaeomagnetic study on the Valaam sill at Lake Ladoga area in north-western Russia, Salminen and Pesonen (2007) showed that at ca. 1.46 Ga Baltica was located at southern latitudes, and was connected with eastern Greenland along its present north-eastern Timanian margin. That reconstruction was based on a mean palaeopole combining the data from the Valaam sill, Salmi volcanic rocks (Shcherbakova et al., 2006) and dolerites (Lubnina et al., 2005), all from Lake Ladoga area, and Tuna dolerites in Sweden (Bylund, 1985). Although otherwise well-defined, the primary nature of the remanence of the Valaam sill could not be verified by palaeomagnetic field tests. The reconstruction is in general agreement with our preliminary studies (Lubnina et al., 2005), which indicated a position of Baltica at low southern latitudes.

In order to further constrain the palaeoposition of Baltica within the Columbia supercontinent, we have studied several mafic rock types at Lake Ladoga region. Preliminary palaeomagnetic and palaeostress data on the dolerite dykes were given in Lubnina et al. (2005) and Shcherbakova et al. (2008), which have considerably been improved in the present study. The rocks of the region have many advantages that can be applied in search of well-defined key poles (cf. Buchan et al., 2000).

First, there exist numerous well preserved mafic magmatic rocks of similar composition, but of various emplacement types, both plutonic and volcanic. This makes it possible to evaluate their relationships with host rocks and to use baked contact tests in order to show the primary origin of the remanent magnetization.

Second, there are several new isotopic ages for these mafic rocks. The U–Pb dating of the Valaam sill has ages of 1459 ± 3 and 1457 ± 2Ma (Rämö et al., 2001, 2005). Sm–Nd dating of the Salmi basalts yields a similar age of 1499 ± 68 Ma (Bogdanov et al., 2003), and herein, we will present new U–Pb dating on baddeleyite for the associated dolerite dykes.

Third, the Lake Ladoga area is situated in the central part of the East European Craton (EEC), at a substantial distance from the Phanerozoic orogenic belts (e.g. the Caledonides and Uralides) that surround this craton and could have had influence on resetting the remanent magnetisation (cf. Shipunov, 1998).

2. Geology

The Pasha-Ladoga graben is situated in the southern part of the so-called Raahe-Ladoga zone that separates the Archaean Karelian craton in the northeast from the Palaeoproterozoic 1.95–1.85 Ga Svecofennian orogen in the southwest (Gaál and Gorbatschev, 1987; Gorbatschev and Bogdanova, 1993; Nironen, 1997; Lahtinen et al., 2005). The formation of the Raahe-Ladoga zone involved several stages of transtension and transpression (Gaál and Gorbatschev, 1987; Morozov et al., 2000), and there was a marked period of reactivation at ca. 1.80–1.75 Ga (Kärki and Laajoki, 1995; Eklund et al., 1998; Korsman et al., 1999; Nironen, 2005 and references therein), i.e. roughly concomitantly with the final amalgamation of the East European Craton by the collision of the Fennoscandian and Volgo-Sarmatian crustal blocks farther in the present south-east (Bogdanova et al., 1996, 2008).

The Pasha-Ladoga structure itself is an asymmetric half-graben with a steep north-eastern flank and a gently inclined south-western boundary. The basement mostly consists of rocks belonging to the Palaeoproterozoic Ladoga Formation and granitoids. Within the half-graben is the so-called Salmi Formation (Fig. 1), an almost 300 m thick sequence of red-bed arkoses, conglomerates, siltstones, tuffs, cross-bedded tuffitic sediments, and basaltic flows, which were in part deposited in a basin associated with the ca. 1.55–1.53 Ga Salmi rapakivi granite massif (Amelin et al., 1997). A swarm of mafic dykes follows the north-eastern flank of the graben (Svetov, 1979; Amantov et al., 1996; Bogdanov et al., 2003), while its southern part is buried beneath a late Proterozoic (Ediacaran) to Phanerozoic platform cover and can only be traced geophysically and by rare drill cores.

On the basis of drill cores and a single outcrop along the Tulemajoki River (site SA in Fig. 1) two sub-formations of the Salmi Formation have been distinguished above the basal conglomerates and red sand- and siltstones (e.g. Svetov, 1979). The lower one consists of arkosic sandstones, quartz sandstones, quartzites and conglomerates that contain pebbles of rapakivi granite and Svecofennian metamorphic rocks. Within this sub-formation there are also nine olivine-basaltic flows, the Salmi basalts, which are separated from each other by interbeds of tuffs and tuffitic siltstone. From one of the basaltic rocks, a Sm–Nd age of 1499 ± 68 Ma has been obtained (Bogdanov et al., 2003). Chemically, the Salmi basalts are rich in Ti, Fe and P, and thus correspond to jotunitic rocks (Nosova, 2007), which indicates their possible relation to the anorthosite-rapakivi magmatism (cf. Duchesne et al., 1999).

The upper Salmi sub-formation consists of red sandstones and gravelstones with conglomerate lenses and tuffs alternating with four- to six thin flows of olivine basalt. The presence of gravelstones and claystones between the two sub-formations indicates a break in the mafic volcanic activity. However, acid dacitic and rhyolitic tuffs are present within these intervening sedimentary rocks.

Up to 200 m thick Valaam dolerite sill in the Valaam archipelago (Fig. 1) was intruded into the Salmi Formation most probably coevally with the uppermost Salmi volcanics (Svetov and Sviridenko, 1995; Amantov et al., 1996; Rämö et al., 2005). The Valaam sill is almost horizontal, the average dip being only 0.2° to the northeast. It is compositionally differentiated and consists of melanocratic, coarse to medium-grained olivine dolerite at the bottom, coarse mesocratic gabbro-dolerite with titanomagnetite in the middle part, and medium- to coarse-grained ophitic gabbro-dolerite with miarolitic voids and pipe vesicles at the top. Monzonitic and quartz syenitic sheet intrusions occur in the middle and upper parts of the sill. They also form dome-like intrusions in some places (Svetov and Sviridenko, 1995).

The Valaam sill features a number of strike-slip faults with a sinistral sense of displacement, defining an “offset-like” structure of the sill. These faults probably developed in the Phanerozoic or even more recently (Svetov and Sviridenko, 1995).

Along the north-north-western shore of Lake Ladoga, there occurs a narrow belt of mafic ferrous dolerite (“sortavalite”) dykes that cross cut both the Salmi rapakivi granites and the gneisses of the host Ladoga Formation (Fig. 1). Even though all these dykes have similar chemical compositions, two sub-groups (types A and B) can structurally be distinguished (Fig. 2A). They have slightly different trends (Fig. 2B), which indicates differing stress fields during their emplacement (Fig. 2C). Type-A dykes strike 320–330° and dip 40–70° NNE, whereas the type-B dykes trend 350–360° and are vertical–subvertical. The extensional stress vectors of the two sub-groups yield two well-defined maxima (Fig. 2C). Both maxima are close in space but, however, the principal extensional and compres-

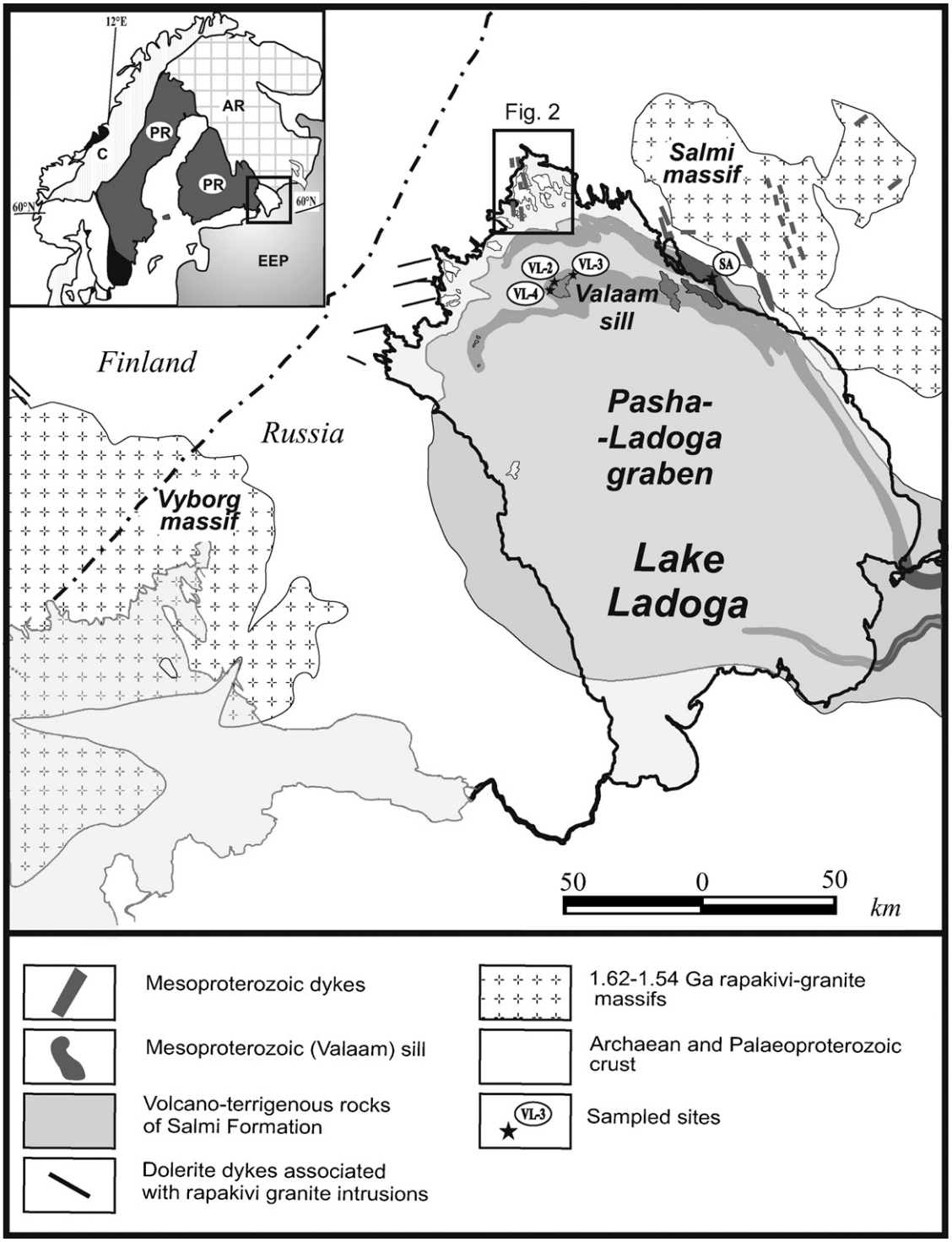


Fig. 1. Geological sketch map of the Lake Ladoga region (South Russian Karelia). Sampled sites—VL-2, VL-3, VL-4: Mesoproterozoic Valaam sill; SA: Salmi basalts. The inset map shows the major tectonic units in the Fennoscandian Shield—AR: Archaean granite–gneiss–greenstone crust, PR: Palaeoproterozoic Svecofennian orogen, C: Caledonides, EEP: East-European Platform. The square shows the present study area.

sional axes of the type-A and type-B dykes have slightly different orientations (ENE 69º, dip 32° and ENE 64°, dip 10°, respectively).

In addition to differences in dyke trends, the type-A dykes and type-B dykes differ in their mode of occurrence. Type-A dykes are aphanitic and occur in the western part of the dyke swarm (Helylä, Tamkhanka and Suur-Haapasaari islands, Fig. 2), where they cut the gneisses of the Ladoga Formation in a NNW-SSE direction. They are mostly short and disconnected, ranging in width from a few centimetres to 10–15 m.

The Helylä dyke is straight, up to 2 m thick and it has narrow (1–2 cm) chilled margins. It is composed of dark-grey, fine-grained aphanitic dolerites which are vitreous at the contacts. The dyke on the Tamkhanka island can be followed for about 500 m. It consists of microporhyritic plagioclase-pyroxene dolerite with 10–15 cm thick vitreous contacts. These contacts are uneven and complicated by thin apophyses of glassy material occurring along the foliation of the host quartz-feldspar gneisses and migmatites. The Suur-Haapasaari dyke is actually a complex dyke-vein system ca. 50 m

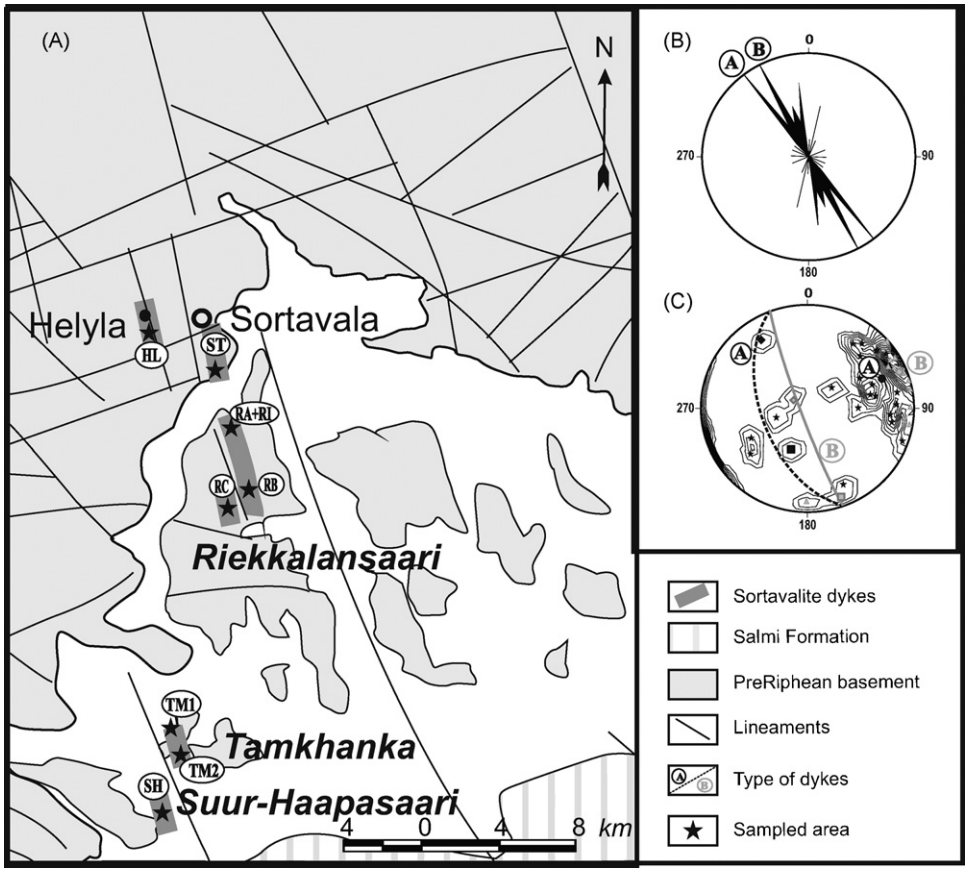


Fig. 2. Generalized geological map of the mafic dykes at the Lake Ladoga region showing the sampling sites of the aphanitic dolerite dykes (A-type) (sites HL, TM1 and TM2 and SH) and sortavalite dykes (B-type) site RA + RI, RB, RC and ST. (A) The map is modified from Svetov (1995); (B) Rose diagram of the predominance strike of dykes; (C) density counter of the extensional stress vectors for dykes (Wulff's projection, low hemisphere). The width of the dykes is shown out of scale.

in length (Vasilieva et al., 2001). It dips ca. 40° to SW, and there are numerous subhorizontal aphanitic/glassy apophyses along the foliation of the host rocks.

Type-B dykes occur on the Riekkalansaari island and also in the city area of Sortavala. The dykes can be followed for about 2 km (Fig. 2). The thicknesses of these dykes are up to 30 m, with 10–15 cm wide chilled, uneven and smoothly curved contacts. The coarseness of this dolerite increases markedly in the central part of the dykes.

3. Samples for the palaeomagnetic and geochronological studies

Altogether 154 oriented samples from eight sites within the Lake Ladoga region were collected. Part of the samples was taken with a water-cooled portable drill and part as block samples. The samples were oriented with magnetic and partly with a sun compass. Samples from different formations were taken as follows:

- Twenty five drill cores were taken from the two flows of the Salmi basalts in two sites along the Tulemajoki river (SA in Fig. 1), which trends perpendicular to the lava flow direction.
- In the well exposed Valaam sill, samples of three different types of rocks were collected at three sites on the Valaam Island (sites VL-2, VL-3 and VL-4 in Fig. 1). Altogether twenty-five drill cores of medium-grained ophitic gabbro-dolerite and monzonite-quartz syenite were taken in the northern part of the island (site VL3 in Fig. 1). Thirty-five drill cores of melanocratic coarse- to medium-grained olivine dolerite were taken in the western part of island (site VL-2 in Fig. 1), and twenty-eight drill cores of mesocratic gabbro-dolerite and syenites were taken in the south-western part (site VL-4 in Fig. 1).
- Four type-A dolerite dykes were sampled at four sites. Ten drill cores and five hand samples were taken near the Helylä village (HL in Fig. 2), nineteen samples from the Suur-Haapasaari island (SH in Fig. 2), and 24 and 40 drill cores were taken from two dykes in the northern and southern parts of the Tamkhanka island, respectively (TM1 and TM2 in Fig. 2). In order to carry out baked contact tests for these dykes, fifteen samples were also taken from the host granitoid gneisses at distances of 5–30 m from the dykes in the northern part of the Tamkhanka island (TM1).
- Four type-B dolerite dykes were sampled at five sites. In total, 33 samples were collected from Riekkalansaari island (sites RA, RB and RC in Fig. 2) and five samples from close to the city of Sortavala (site ST in Fig. 2).

In order to determine exact ages of the dolerite dykes, a fresh medium-grained ophitic dolerite was taken from the central part of the type-B dyke on the Riekkalansaari island at site RA + RI (Fig. 2). Sample RI-1 contains baddeleyite, which was used for U–Pb dating.

4. Methods

4.1. Palaeomagnetic studies

The drill core samples were cut into standard cylinders and the block samples were sawed into cubes of 2 × 2 × 2 cm in size. Part of the sample suite was analysed at the palaeomagnetic laboratory of Geological Survey of Finland (GTK) and part at the palaeomagnetic laboratory of VSEGEI, St. Petersburg, Russia. Remanence measure-

ments were performed using 2G cryogenic magnetometer (GTK) and a JR-5A spinner-magnetometer (VSEGEI). Conventional progressive thermal or alternating field (AF) demagnetizations was applied to all specimens. Chemical changes during thermal cleaning were monitored by measuring the magnetic susceptibility after each heating step using a KLY-2 Kappa bridge.

NRM components were visually identified by using stereographic and orthogonal projections (Zijderveld, 1967). The directions of components were calculated by principal component analysis method (Kirschvink, 1980). Mean directions were calculated according to Fisher (1953). All calculations and graphic representations of the results were performed using software by Enkin (1994), Leino (1991) and Torsvik and Smethurst (1999).

4.2. Rock magnetic studies

Thermomagnetic measurements (magnetic susceptibility versus temperature, K/T curves) were carried out for one sample from each of the four formations (Salmi basalts, Valaam sill, type-A and type-B dykes). The Curie point was determined using the maximum in the second derivative of the thermomagnetic curve, at the point of maximum curvature in the thermomagnetic curve which is a reasonable estimate of the Curie point (e.g. Tauxe, 2002). Each Curie temperature shown in the K/T curve characterizes a ferromagnetic mineral or phase in the rock. The possible differences in heating and cooling curves indicate chemical changes in minerals during the thermal treatments.

In order to further define the magnetic mineralogy of the dykes, three component IRM and subsequent thermal demagnetizations, the Lowrie tests (Lowrie, 1990), were carried out for representative specimens from three Riekkolansaari dykes and on one sortavalite dyke. In the Lowrie test the minerals are identified based on their remanent coercivities and unblocking temperatures. The specimens were first AF demagnetized at 160 mT. After imparting the IRM along the *z*-axis up to the highest field of 1.5 T, the magnetization was then imparted along the *y*-axis in a magnetizing field of 0.4 T. Then, the samples were subjected to the magnetizing field of 0.12 T along the *x*-axis. After acquisition of IRM along the three orthogonal axes, the samples were thermally demagnetized in 17 steps in the temperatures between 75 °C and 620 °C. Intensity curves of each axis were produced separately and the magnetic minerals were determined based on the unblocking temperatures.

Lowrie–Fuller tests (Lowrie and Fuller, 1971) were carried out on specimens from three Riekkalansaari dykes and on one sortavalite dyke in order to study the magnetic grain sizes of the magnetic minerals. The specimens were first stepwise demagnetized up to peak alternating field of 160 mT. Anhysteretic remanent magnetization (ARM) was then produced in a peak AF of 120 mT combined with a direct field (DC) of 0.075 mT. The ARM was then stepwise demagnetized at the peak AF level of 100 mT. The samples were then given a saturation isothermal remanent magnetization (SIRM) with a field of 1.5 T for the Riekkalansaari samples and 1 T for the sortavalite sample. SIRM was then stepwise demagnetized with a peak AF level of 100 mT.

Other rock magnetic studies (strong field thermomagnetic curves, hysteresis measurements shown in Day plots and thermomagnetic DS tests) as well as SEM studies for part of the dolerite dykes are shown also in the recent paper by Shcherbakova et al. (2008).

4.3. Baddeleyite U–Pb geochronology

Approximately 1 kg of the sample RI-1 from the type-B dolerite dyke was processed. The separation of baddeleyite followed the procedures described by Söderlund and Johansson (2002). About 90 grains and fragments of fresh, dark brown, baddeleyite were recovered with an average size of ca. 40 μm. Dissolution, ion exchange chromatography and mass spectrometry were performed at the Laboratory of Isotope Geology (LIG) at the Swedish Museum of Natural History in Stockholm. Between 10 and 18 fragments/grains were combined in each analysis. The baddeleyite grains were transferred to mini teflon capsules and washed in 2.5 M HNO_3 on a hot plate for ca. 15 min. The samples were thereafter washed several times in ultra-clean H_2O. ^{205}Pb–$^{233–236}U$ tracer and 10 drops of HF and HNO_3 (10:1) were added to the capsules. The baddeleyite grains were completely dissolved over night at 210 °C. After evaporation on hot plate the sample was re-dissolved in 3.1 M HCl and loaded on 50 μl columns with anion exchange resin. After washing by 3.1 M HCl, uranium and lead were eluted by H_2O and dried on a hot plate. The sample was loaded on a single Re filament together with silica gel. Mass spectrometry analysis was performed on a Finnigan MAT 261. ^{208}Pb, ^{207}Pb, ^{206}Pb and ^{205}Pb were measured employing Faraday collectors. The $^{207}Pb/^{204}Pb$ used for common lead correction was measured in dynamic collector mode using a Secondary Electron Multiplier. After completing Pb isotopic measurements in the 1180–1300 °C range, the U-isotopic composition was analyzed in dynamic multicollector mode at a filament temperature in excess of 1350 °C.

The total procedural blank is estimated to about 2 pg Pb and <1 pg U. Mass fractionation is 0.1% per mass unit for Pb, determined by replicate analyses of NBS standards SRM 981 and SRM 983. U fractionation was determined directly from the measured $^{233}U/^{236}U$ isotopic ratio. Initial Pb compositions were taken from the model of Stacey and Kramers (1975) at 1450 Ma. The Pb blank composition used was $^{206}Pb/^{204}Pb = 18.5$, $^{207}Pb/^{204}Pb = 15.6$, and $^{208}Pb/^{204}Pb = 38.5$. The decay constants for ^{238}U (1.55125×10^{-11}) and ^{235}U (9.8485×10^{-10}) are those recommended by Steiger and Jäger (1977). The listed uncertainties of the Pb/U ratios (Table 1) were calculated by propagating the within-run error for measured isotopic ratios and the uncertainties in fractionation (±0.04 for Pb, absolute uncertainties) in Pb and U blank concentrations (±50%), and in the Pb blank composition (2% for $^{206}Pb/^{204}Pb$ and 0.2% for $^{207}Pb/^{204}Pb$).

5. Results

5.1. U–Pb results

The U–Pb data are given in Table 1 and shown in a Concordia plot in Fig. 3. Three of the four fractions are linearly distributed relatively close to the Concordia trajectory at 1452 ± 12 Ma (2 s) with an acceptable MSWD of 1. The choice of the lower intercept is based on the assumption of a Caledonian imprint, as indicated from apatite-fission track data from areas near the Bothnian Sea (e.g. Larson et al., 1999; Murrell, 2003). The fourth fraction plots slightly to the right of the discordia. Despite careful washing and the low risk of contamination when using the separation technique employed, we cannot exclude that the slightly higher $^{207}Pb/^{206}Pb$ age for this analysis is indeed due to sample contamination.

5.2. Rock magnetic results

In thermomagnetic measurements the heating and cooling curves show Curie temperatures from 580 to 600 °C indicating the presence of Ti-poor titanomagnetite or pure magnetite. All samples (Fig. 4) show an increase in susceptibility in the heating curve just before the Curie temperature (a Hopkinson peak). This is the characteristic behavior of single domain (SD)- and pseudosingle-domain (PSD) magnetite grains (Dunlop, 1983).

Table 1
U–Pb TIMS data.

Sample	Mineral type[a]	U/Th	Pbc/ Pbtot[b]	$^{206}Pb/^{204}Pb$	$^{207}Pb/^{235}U$	±2s % err	$^{206}Pb/^{238}U$	±2s % err	$^{207}Pb/^{235}U$	$^{206}Pb/^{238}U$	$^{207}Pb/^{206}Pb$	±2s % err	Concordance (%)
				raw[c]	[corr][d]				[age, Ma]				
SK	Bd1	6.7	0.017	1910	3.2446	0.50	0.25389	0.29	1467.9	1458.5	1481.5	7.5	97.9
	Bd2	9.0	0.057	693	3.0289	0.84	0.24148	0.56	1414.9	1394.4	1446.0	11.6	94.6
	Bd3	11.2	0.022	1066	3.0926	0.90	0.24737	0.75	1430.9	1424.9	1439.8	10.0	98.3
	Bd4	4.5	0.009	1013	2.9324	1.03	0.23527	0.96	1390.3	1362.0	1433.9	8.0	94.3

[a] Bd = baddeleyite, 10–18 grains/fragments were combined in each analysis.
[b] Pbc = common Pb; Pbtot = total Pb (radiogenic + blank + initial).
[c] Measured ratio, corrected for fractionation and spike.
[d] Isotopic ratios corrected for fractionation (0.1% per amu for Pb), spike contribution, blank (2–5 pg Pb and <1 pg U), and initial common Pb. Initial common Pb corrected with isotopic compositions from the model of Stacey and Kramers (1975) at 1450 Ma.

Minor differences between heating and cooling curves of the samples (cf. sample RI-15) suggest that during the heating some mineralogical changes have taken place.

The Lowrie tests (Fig. 5) for the Riekkalansaari dykes show that in all samples the magnetization of the dominant medium coercivity fraction decreases at 500–580 °C, suggesting that SD magnetite is the main carrier of remanence. Specimen RC4-3A from the Riekkalansaari dyke RC also shows a smaller drop of intensity near the temperature of 300–400 °C, probably related to a minor amount of pyrrhotite. The hard and soft fractions are very weak compared to the intermediate fraction. They are totally demagnetized at ca. 570 °C indicating magnetite as the remanence carrier.

The Lowrie–Fuller tests (Fig. 6) for the Riekkalansaari dykes indicate that ARM is stronger than SIRM, further suggesting that the remanence is carried by SD/PSD magnetite. All the rock magnetic results thus point out that the dykes are capable of carrying hard coercivity remanence that is least vulnerable to later geological effects.

5.3. Palaeomagnetic results

Palaeomagnetic results from the studied formations are shown in Table 2. Examples of demagnetization behaviours are shown in Figs. 7, 8, 10, 12 and 13) and summaries of the remanent magnetization components are shown in Figs. 9 and 14.

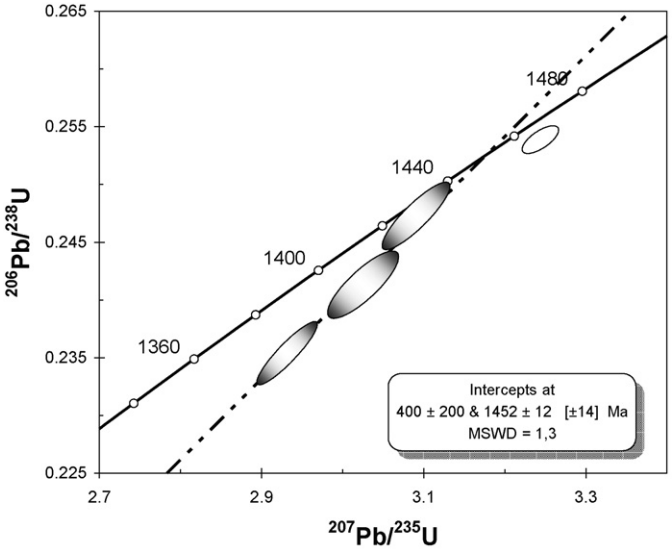


Fig. 3. Concordia plots of U–Pb baddeleyite data of the Riekkalansaari dolerites (sample RA-1). Error ellipses indicate 2σ error.

5.3.1. Salmi basalts

During stepwise thermal and alternating field (AF) demagnetization experiments, two components of NRM were isolated in the majority of specimens. Component I_S is of single polarity and is directed to the northeast with shallow positive (downwards) inclination. It is isolated in the temperature interval of 380–440 °C or in alternating fields below 40 mT. The second remanence component, H_S with dual polarity was demagnetized in high temperatures up to 540–580 °C and AF fields above 40 mT. The normal polarity component H_N points to the NE with negative shallow inclination (for example, sample TL 9-33, Fig. 7), and the reversed polarity component H_R points to the SW with positive inclination (sample TL 9-17, Fig. 7). Sample TL9-10 (Fig. 7) is an example of a case where component H_R occurs as a stable single component. Most samples of the Salmi basalt carry the reversed polarity H_R component. The normal and reversed directions are antipodal (Fig. 14) and pass the reversal test (McFadden and McElhinny, 1990) with classification C ($\gamma = 17.63°$ and $\gamma_c = 19.72°$). Based on the positive reversal test, this component is interpreted to be of primary origin. For the mean directions of this component, see Table 2 and Fig. 14.

5.3.2. Valaam sill

Both thermal and AF demagnetization were used in isolating the components of the Valaam sill. Three components of NRM were isolated in the majority of specimens. At sites VL2 and VL3 a low coercivity and low unblocking temperature component I_V, that points to the NE with shallow positive (downward) inclinations, was isolated in the temperature interval of 300–500 °C and below AF fields of 40 mT. A similar component was isolated sporadically also at site VL4 (Fig. 8). This component has single polarity and it compares well with the I_S component of the Salmi basalt. The second component M was isolated almost in the same temperature and AF fields as component I_V, but it points more to the E-NE and has an intermediate inclination (Table 2). Above 540 °C and 40–100 mT, a third, high coercivity/temperature H_V-component with a northeast pointing shallow upward directed inclination was identified at all sites VL2, VL3 and VL4. The remanence corresponds with that of the H_S component of the Salmi basalt. Examples of AF and thermal demagnetization of the samples are given in Fig. 8. Both in the syenites and dolerites of the Valaam sill, a few samples show antipodal H_V directions (Table 2). The remanence resides in magnetite. The site-mean remanence directions for the H_N component are well clustered. The directions and corresponding palaeomagnetic poles averaged for all sites are shown in Table 2 and Fig. 9.

5.3.3. Type-A dykes

The samples of the Helylä dolerite dykes (HL) were more effectively demagnetized by thermal demagnetization (Fig. 10), whereas the samples from the Tamkhanka (TM) and Suur-Haapasaari

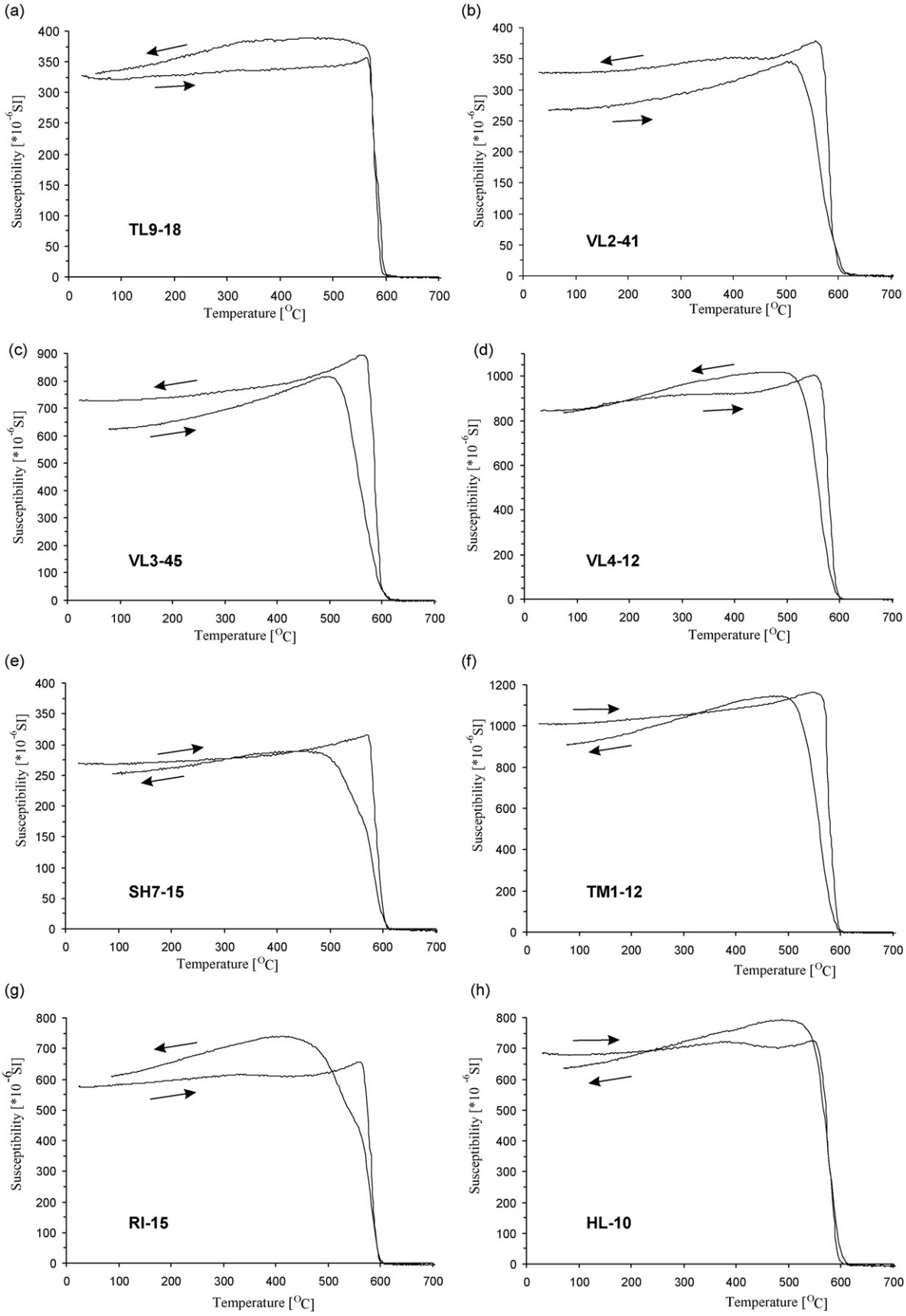


Fig. 4. Thermomagnetic curves (susceptibility vs. temperature) for specimens of the magmatic complexes from the Lake Ladoga region: (a) Salmi basalts (sample TL9-18), (b–d) Valaam sill (b, sample VL2-41, site VL-2; c, sample VL3-45, site 3; d, sample VL4-12, site 4), (e) Type-A dyke from the Suur-Hapasaari island (sample SH7-15), (f) Type-A dyke from the Tamkhanka island (sample TM1-12), (g) Type-B dyke from the Riekkalansaari dyke (sample RI-15) and (h) Type-A dyke from the Helylä (sample HL-10).

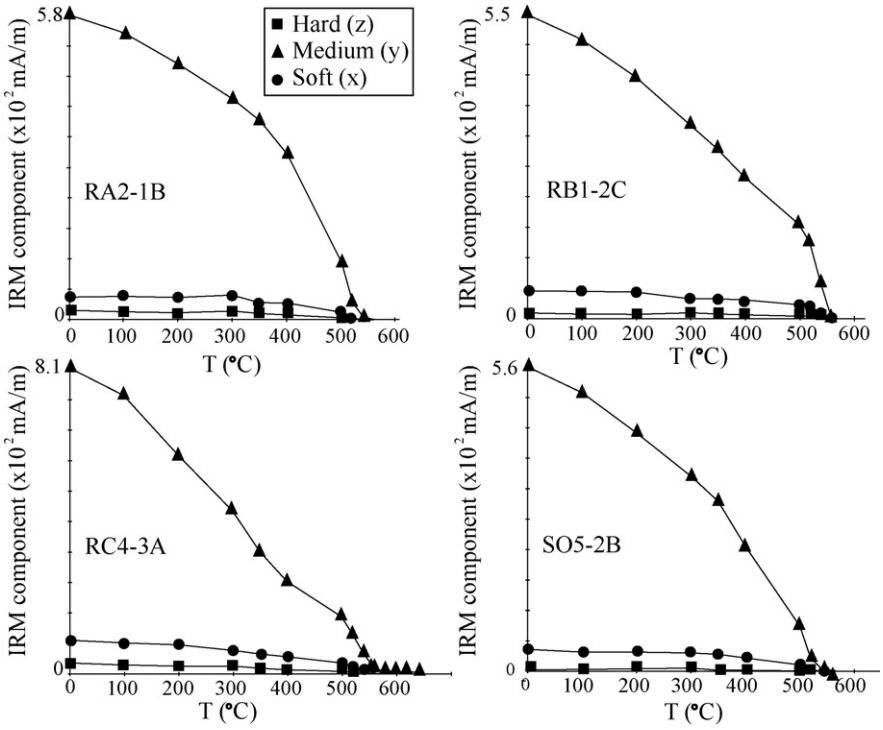


Fig. 5. Examples of thermal demagnetization of three-orthogonal IRM in the Riekkalansaari (RA, RB and RC) and sortavalite (ST) dykes. Hard (*z*) 1.5–0.4 T, intermediate (*y*) 0.4–0.12 T, and soft (*x*) <0.12 T fractions.

(SH) dykes were more effectively demagnetized by the AF method.

Thermal demagnetization of dykes HL and SH indicates the presence of two magnetic phases that, based on unblocking temperatures, are probably low-titanium titanomagnetite or magnetite (demagnetized between 520 and 590 °C) and titanomaghemite (unblocking temperatures 330 and 370 °C). In central parts of these dykes the titanomaghemite component has a NE declination and intermediate positive inclination, corresponding to the mean direction of I_S and I_V-components of Salmi basalts and the Valaam sill (Table 2). In the marginal parts of the type-A dykes the remanence is unblocked within the whole temperature range of 330–590 °C and shows extremely stable behaviour with a single component, H_A (Fig. 10). The occurrence of this highly stable remanence component especially in the chilled margins of the dykes is evidence that the magnetization took place during cooling of the dykes. This remanence component has a north-easterly declination and shallow negative inclination (Table 2, Figs. 10 and 14) which is close to the directions of the H_S and H_V components of the Salmi basalts and Valaam sill, respectively. The remanence directions of separate type-A dykes are somewhat different as dykes HL and TM1 have slightly lower inclinations than dykes SH and TM2. It is believed that the characteristic component H_A of the type-A dykes represents the primary remanence of the dykes. In order to verify this, a baked contact test was carried out for the Tamkhanka dyke TM1.

5.3.4. Baked contact test

Fig. 11 shows the location of the samples for the baked contact test of type-A aphanitic dykes in the northern part of the Tamkhanka island (TM1). Sample TM-1/4 (Fig. 12) taken from the dyke margin yields a shallow NE directed, single stable remanent magnetization component. Sample TM-1/25 taken from the baked granite-gneiss of the Ladoga Formation has a similar NE

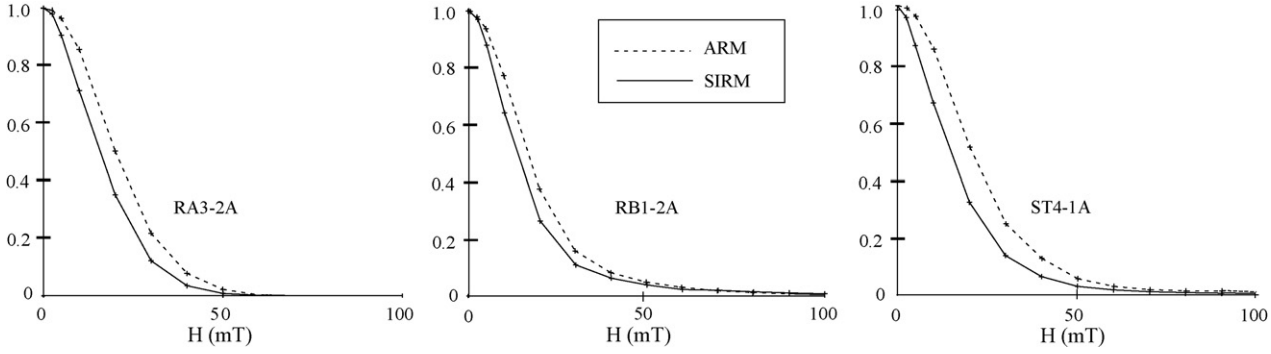


Fig. 6. Lowrie–Fuller tests for samples from the Riekkalansaari dykes RA, RB and from the sortavalite dyke ST. Stippled curves denote AF demagnetization of ARM and solid curves of SIRM.

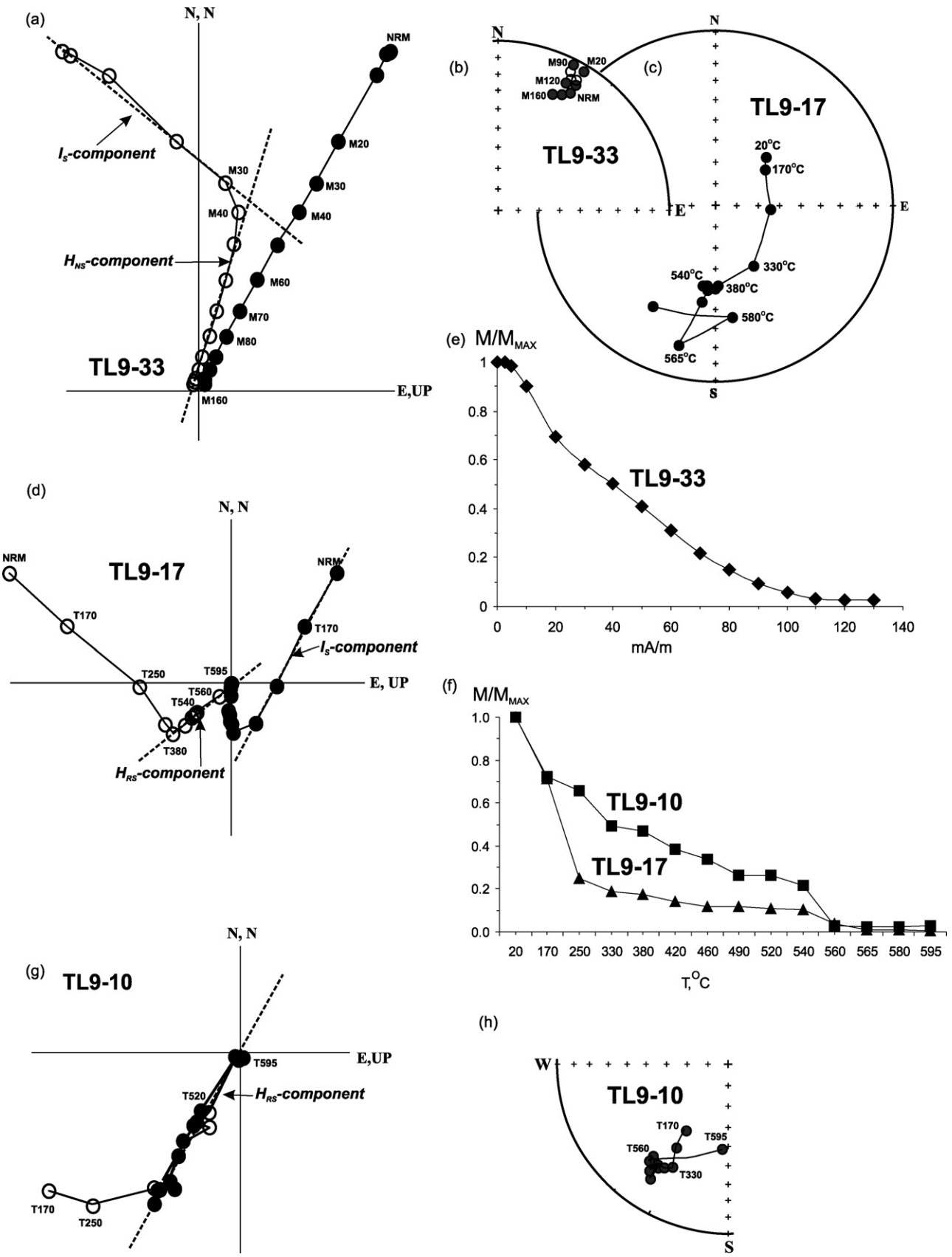


Fig. 7. Examples of thermal and AF demagnetization behavior of specimens from the Salmi basalt. (a, d, g) orthogonal vector projections, (b, c, h) stereographic projections, (e) relative NRM intensity decay curves upon AF demagnetization, (f) upon thermal demagnetization. Open (closed) symbols denote projections onto vertical (horizontal) planes. Numbers at demagnetization steps denote peak alternating field (M, mT) or temperature (T, °C).

Table 2
Palaeomagnetic results for the magmatic complexes of the Lake Ladoga formation.

Component	B/N	Dec (°)	Inc (°)	α95 (°)	*K*	Plat (°N)	Plong (°E)	dp (°)	dm (°)	*A*95
Salmi Basalt										
I_S	2/15	36.3	44.3	5.9	42.4	−47.0	344.6	7.4	4.7	5.9
H_R	1/22	212.5	13.6	7.1	20.1	17.2	178.0	7.3	3.7	5.2
H_N	1/3	47.5	−24.1	19.0	43.1	7.1	165.0	20.3	10.9	14.9
Mean Hs	2/25	214.1	15.5	6.7	19.5	15.8	176.6	6.9	3.5	4.9
Valaam Sill										
Site VL2										
I_{V2}	2/9	36.0	17.7	6.9	56.4	−31.4	348.1	7.2	3.7	5.1
M_N	1/7	75.6	48.0	10.4	34.4	−32.0	298.1	13.6	8.9	11.0
H_{V2}	2/14	30.9	−11.0	4.3	85.8	−18.9	178.2	4.4	2.2	3.1
Site VL3										
I_{V3}	3/20	32.4	56.0	6.5	26.1	−57.8	337.0	9.7	6.7	7.9
H_{V3}	3/21	32.0	−10.4	4.5	49.9	18.7	183.0	4.6	2.3	3.2
Site VL4										
I_{V4}	2/12	27.0	53.3	8.3	28.2	−57.5	346.4	11.5	8.0	9.6
H_{V4}	2/14	25.8	−11.0	5.6	51.3	20.1	183.4	5.7	2.9	4.0
Mean Valaam sill										
I_V	*3/41	32.1	47.0	6.0	14.6	−50.6	343.4	7.8	5.0	6.2
H_V	*3/49	29.9	−10.8	2.7	56.1	19.3	179.2	2.7	1.4	1.9
Mean H_V	*3/49	29.6	−10.8	5.0	615.0	19.3	179.2	5.1	2.6	3.6
Type-A Dykes										
Suur-Haapasaari (SH)										
I_{SH}	3/36	33.3	50.4	7.5	11.2	−52.7	339.9	10.1	6.8	8.3
H_{SH}	3/59	40.0	−26.1	4.4	58.3	8.3	171.6	4.3	2.3	3.2
Helylä (HL)										
I_{HL}	1/10	25.2	34.9	14.7	11.7	−44.2	356.6	16.9	9.7	12.8
H_{HL}	1/11	45.0	−16.9	9.9	22.3	11.6	165.2	10.2	5.3	7.4
Tamhanka 1 (TM1)										
I_{TM1}	2/32	36.9	35.4	7.8	11.6	−40.8	342.3	9.0	5.2	6.9
H_{TM1}	2/56	31.8	−17.1	4.3	48.3	15.4	178.0	4.4	2.3	3.2
Tamhanka 2 (TM2)										
I_{TM2}	2/33	20.4	44.1	6.7	14.7	−51.8	0.2	8.4	5.3	6.6
H_{TM2}	2/40	37.2	−24.7	4.6	25.1	9.9	174.0	4.9	2.6	3.6
Mean Type-A dykes										
I_A	*4/111	29.0	41.4	10.7	75.2	−47.2	349.7	13.1	8.0	10.2
Mean H_A	*4*/166	38.2	−20.0	6.4	142.1	12.1	172.2	6.7	3.5	4.8
Type-B Dykes										
RA + RI										
I_{RI}	2/21	28.2	46.2	7.1	21.2	−51.2	348.5	9.1	5.8	7.3
H_{RI}	1/5	202.8	29.7	15.5	25.2	10.5	188.4	17.2	9.5	12.8
H_{NI}	1/19	29.9	−21.1	6.0	32.0	13.9	180.4	6.3	3.3	4.6
Mean RA + RI	2/24	28.6	−23.3	5.6	28.9	14.4	182.2	5.8	3.0	4.2
RB	1/5	25.5	−18.2	5.4	117.8	16.2	184.6	3.8	7.4	7.1
RC	1/4	30.8	−19.9	7.1	170.8	14.1	179.5	3.9	7.4	6.6
Sortavala (ST)	1/5	37.3	−15.0	7.3	68.0	14.8	172.4	4.9	9.6	9.4
Mean H_B	*4/38	28.3	−20.5	5.5	504.2	14.9	182.1	3.0	5.8	4.1
Grand Mean H	*4/12/278	32.5	−16.7	7.1	168.6	15.2	177.1	7.3	3.8	5.5

Note: B/N, number of sites/samples, used for mean calculations; *Statistical level used for mean calculation. Dec and Inc are the mean declination and inclination respectively; α95 is the radius of the circle of 95% confidence; *K* is the Fisher's (1953) precision parameter of the mean pole; Plat and Plong are the palaeolatitude and palaeolongitude for the Virtual Geomagnetic Poles, dp and dm are the semi-axes of the oval of 95% confidence; *A*95 is the radius of the circle of 95% confidence of the mean pole.
Grand Mean H Pole calculated for coordinates: Glat = 61.7°N Glong = 30.5°E.

pointing high coercivity remanence direction to the dyke (Fig. 12). Samples TM-1/37-1/39 from the unbaked gneisses yields two directions. In high AF fields the remanence has a NW pointing negative inclination direction (Fig. 12, specimen TM-1/39), which is clearly different from the previous NE pointing direction. The result thus indicates that the baked contact test is positive and the NE pointing shallow inclination remanence direction is the primary remanent magnetization of the A-type aphanitic dyke. In the baked and unbaked gneisses, we additionally obtained in low AF fields a similar moderately NE pointing positive inclination direction as the I_A-direction in the samples from the dykes (Fig. 12). Based on a negative contact test for this low coercivity component, it is implied that this component represents a secondary magnetization.

5.3.5. Type-B dykes

The samples of type-B dykes from the Riekkalansaari and Sortavala can be subdivided into weakly (NRM 0.7–1.5 A/m) and strongly (NRM 2.5–4.3 A/m) magnetic types. The contact zones of the dykes carry mostly a single-component NRM, component

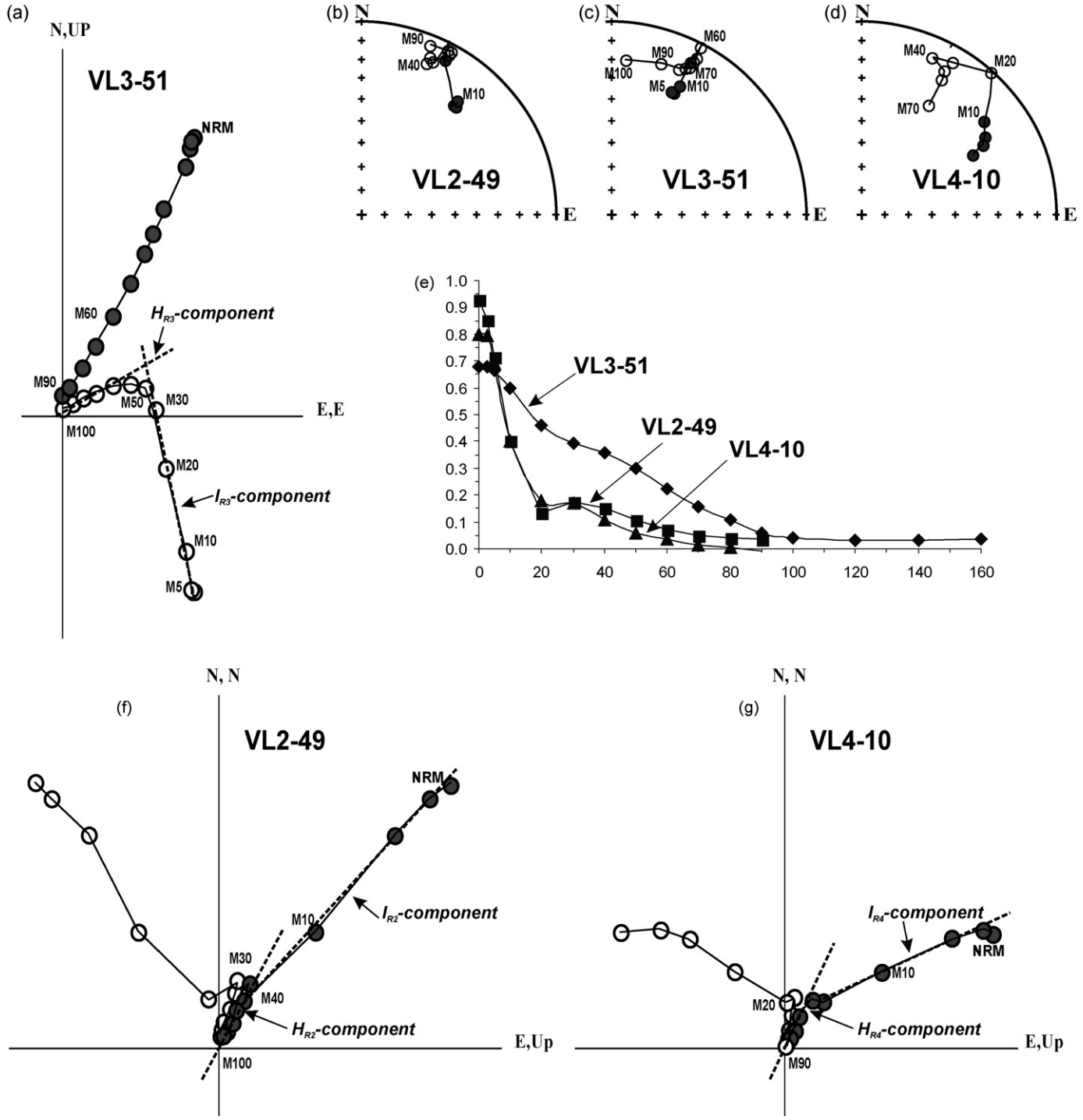


Fig. 8. Examples of AF demagnetization behavior of specimens from different sites (VL2, VL3, VL4) of the Valaam sill. (a, f, g) orthogonal vector projections, (b–d) Stereographic projections, (e) relative NRM intensity decay curves upon AF demagnetization. For explanations, see Fig. 7.

H_B, which unblocks gradually below 595 °C and above 40 mT. It has a shallow NE pointing negative inclination direction (Fig. 13) corresponding to the remanence of component H_A of the type-A dykes. The principal carrier of this component is magnetite (Figs. 4 and 5). Samples from the central part of the RA + RI dyke show an antipodal direction H_R (Table 2, Figs. 13 and 14). Some specimens from the central part of the RA + RI dyke demonstrate a self-reversal effect, which may indirectly support a primary origin of this component. The reversal test of McFadden and McElhinny (1990) for the H_R and H_N components is positive with classification C (i.e. observed angle between the directions is $\gamma = 10.72°$; critical angle $\gamma_c = 13.43°$). This further suggests a primary origin for that component. Samples from the central part of the dykes have multicomponent NRM. A low-stability remanence component I was removed between temperatures of 200 and 440 °C and below AF fields of 20–40 mT. It is directed NE and downwards corresponding with the *I*-direction obtained in the other formations (Fig. 13).

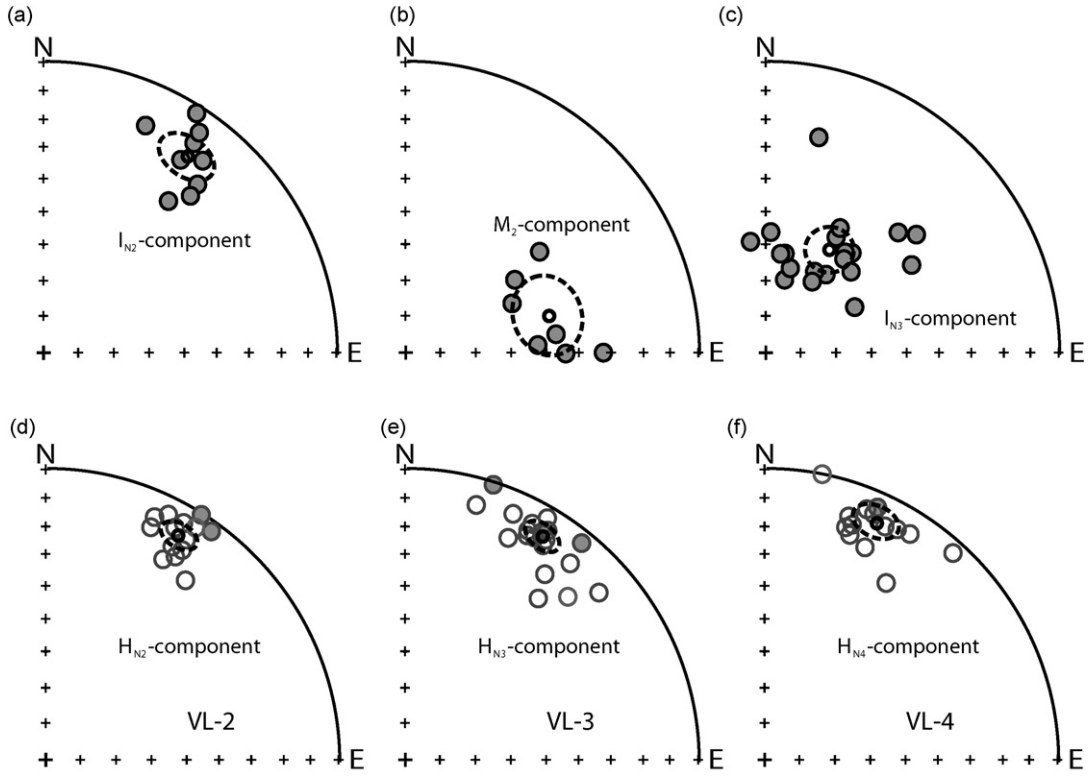


Fig. 9. Sample mean palaeomagnetic directions for different components of the Valaam sill. The mean directions of sites are shown as smaller circles with α95 confidence cones about the means. (a and b) Low coercivity and low unblocking components I_{N2} and M_2 from site VL2, (c) low coercivity and low unblocking component I_{N3} from site VL3, (d–f) high coercivity and high unblocking component H from sites VL2 (H_{N2}), VL3 (H_{N3}) and VL4 (H_{N4}).

6. Discussion

6.1. Age correlations

The new U–Pb baddeleyite age of 1452 ± 12 Ma for the Fe-rich olivine dolerites of the type-B dyke swarm correlates within error limits with the age of the Valaam sill, dated at 1459 ± 3 and 1457 ± 2 Ma (Rämö et al., 2001, 2005). The new age also links the dolerite rocks at Lake Ladoga area to the 1.47–1.44 Ga mafic dykes and sills abundant within the Palaeoproterozoic crust in Dalarna (Central Sweden), and in the Sveconorwegian orogen in south-western Fennoscandia. Brander and Söderlund (2008) have pointed out a compositional trend of the 1.47–1.44 Ga magmatism from north to south. South of a tentatively drawn E-W trending line (broadly across the large lakes in southern Sweden) magmatism of this age is characterised by relatively small isolated granite plutons, for instance on the island of Bornholm, in Blekinge, SW Scania and in the Baltic states. North of this line, the magmatism is exclusively of mafic composition (basalts and dolerite sills and dykes), often associated with local basins and rift sediments. Based on the new ages of the olivine dolerites, the mafic intrusions in the Lake Ladoga area in NW Russia can be tentatively associated with those in Central Sweden.

6.2. The poles

Site-mean palaeomagnetic poles for the mafic rocks of the Lake Ladoga region were calculated from the remanent magnetization data of each formation (Table 2). As seen in Table 2, the poles of the high coercivity and high temperature component H for different rock types correspond to each other, suggesting that the remanence was acquired nearly concomitantly in all formations.

Positive baked contact test for the type-A dolerite dyke TM1 of the Tamkhanka island implies that the remanent magnetization of component H is primary. Furthermore, in the type-B Riekkalansaari dykes and in the Salmi basalts, the H component has both normal and reversed polarities (Table 2), and positive reversal tests for these two different mafic rock types indicate a primary origin for the H component. Based on the new U–Pb dating of the Riekkalansaari dyke we conclude that the magnetization was acquired at ca. 1.45 Ga. Secular variation is averaged out since the calculated key pole is obtained from 13 VGP poles for four different lithologies of about the same age. The mean H pole (pole LA) is shown in Table 3 and in Fig. 15. The mean pole I of the low coercivity/temperature component is shown in Fig. 16.

The new palaeomagnetic data from Lake Ladoga formations correspond to the previous similar aged palaeomagnetic results from the Tuna dykes (Bylund, 1985) and Bornholm granites (Lubnina et al., 2009) in Sweden (Table 3). The obtained poles are in-between those from the older Gustaf porphyries in Central Sweden and from the slightly younger (Ernst et al., 2006) Mashak dykes in Southern Urals (Lubnina, 2009b). However, the palaeomagnetic pole from the Valaam sill of the present study (pole H_V, Table 2) is ca. 15° to the east from that obtained by Salminen and Pesonen (2007), and the pole of the present study from the Salmi basalts (H_S) is ca. 30° to the west from the pole by Shcherbakova et al. (2006). Since the Valaam sill is strongly affected by faulting with a left-shear component (Svetov and Sviridenko, 1995), a possible reason for the former deviation may be structural tilting. The difference of the data from the Salmi basalts (Shcherbakova et al., 2006) may be caused by transpressional-to-transtensional deformation along the north-eastern flank of the Pasha-Ladoga graben (cf. Morozov, 1999). Alternatively, the difference may be caused by secular varia-

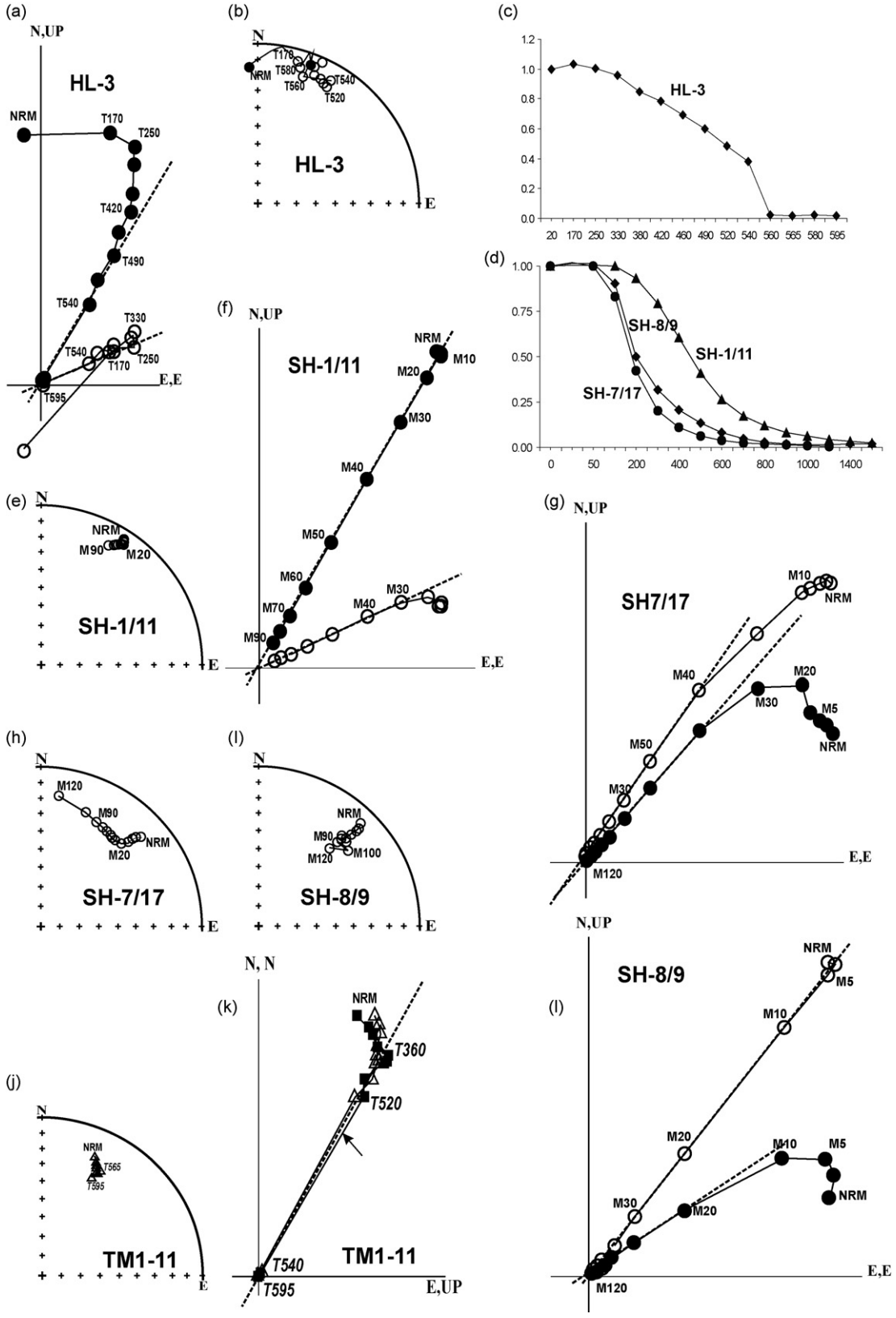


Fig. 10. Examples of thermal and AF demagnetization behaviors of specimens from the aphanitic type-A dolerite dykes (HL Helylä, SH Suur-Haapasaari, TM Tamkhanka). (a, f, g, k, and l) orthogonal vector projections, (b, e, h, i, and j) stereographic projections, (c) relative NRM intensity decay curve upon thermal demagnetization, (d) relative NRM intensity decay curves upon AF demagnetization. For explanations, see Fig. 7.

Table 3
Palaeomagnetic poles, used for proposed reconstructions.

PN	Age (Ma)	Pole	Formation	Dec (°)	Inc (°)	Plat (°)	Plong (°)	A95	Ref.
East European Craton (Baltica)									
1	**1780**	**VZH**	**Vazhinka River fm**	**350**	**31**	**42**	**221**	**7**	**1**
2	1499 ± 68	SA	Salmi basalt	8	−36	6	200	7	2
3	1475 ± 4	GF	Gustaf porphyrites	11	−11	24	184	7	3
4	1461 ± 3	TU	Tuna dykes	26	−5	21	180	7	4
5	1460	BH	Bornholm granites	14	−39	12	182	6	5
6	1458 ± 4	VAL	Valaam sill	41	−11	14	166	2	6
7	**1452 ± 12**	**LA**	**Lake Ladoga region**	**33**	**−17**	**15**	**177**	**5.5**	**This work**
8	**1460**	**MPR**	**Mean 3–7**	**–**	**–**	**17**	**178**	**8**	**This work**
9	**1384 ± 2**	**MS**	**Mashak Fm**	**51**	**−39**	**2**	**192**	**13**	**7**
10	**1265**	**PJI**	**Post Jotnian intrusions**	**51**	**−24**	**4**	**158**	**4**	**8**
Laurentia Craton									
11	**1740 ± 5**	**CL**	**Cleaver Lake dykes**	**197**	**60**	**19**	**263**	**6**	**9**
12	1476 ± 16	SF	St. Francois Mnt	234	9	−13	219	7	10
13	1460 ± 5	MI	Michikamau Intrusion	240	27	−2	218	5	11
14	1450 ± 5	HL	Harp Lake complex	253	23	2	206	4	8
15	1448 ± 49	TR	Tobacco Root Mo	225	62	9	216	10	12
16	**1450**	**MLR**	**Mean 12–15**			**3**	**213**	**13**	**This work**
17	**1267 ± 2**	**MD**	**Mackenzie dykes**	**268**	**12**	**4**	**190**	**5**	**13**
Siberia Craton									
	1473 ± 24	**SG**	**Sololi Group**	**–**	**–**	**34**	**253**	**10**	**14**

Notes: (1) Pisarevsky and Sokolov (2001); (2) Shcherbakova et al. (2006); (3) Lubnina (2009a); (4) Bylund (1985); (5) Lubnina et al. (2009); (6) Salminen and Pesonen (2007); (7) Lubnina (2009b); (8) Buchan et al. (2000); (9) Irving et al. (2004); (10) Meert and Stuckey (2002); (11) Emslie et al. (1976); (12) Harlan et al. (2008); (13) Buchan and Halls (1990); (14) Wingate et al. (2009).

tion, which has not been averaged out in the work by Shcherbakova et al. (2006).

The poles in Fig. 16 have been calculated from both low coercivity and intermediate unblocking temperature component I in the studied mafic rocks. Baked contact tests for this remanence are negative as the studied mafic rocks and the host Palaeoproterozoic gneisses carry a similar low coercivity component. It implies that the magnetization reflects an overall partial reheating event in the area. The pole for the I component (Fig. 16) plots close to the Late Palaeozoic part of the APWP of Baltica (Smethurst et al., 1998) and indicates an age of ca. 300–260 Ma. We suggest that it may reflect geological processes related to the formation/break up of supercontinent Pangaea (Preeden et al., 2009). The origin of I component is not discussed here any further.

6.3. 1450-Ma key pole for the East European Craton

The palaeomagnetic results from the mafic rocks in the Lake Ladoga region suggest that the pole can be regarded as a new 1.45 Ga key pole for the East European Craton. It fully satisfies the reliability criteria for a key pole (Buchan et al., 2000), which include:

- age uncertainty of <±20 Ma,
- positive baked contact and reversal tests,
- the primary remanence is similar for coeval rocks, which were not subjected to regional metamorphism.

Based on the well-defined and precisely dated key palaeomagnetic pole of this study (pole LA) together with the previous poles from the Valaam sill (Salminen and Pesonen, 2007), Tuna dykes (Bylund, 1985; Lubnina, 2009a,b; Lubnina et al., 2007) and Bornholm granites (Bylund, 1985), we propose a new mean Mesoproterozoic key pole (pole MPR, Table 3, Fig. 15) of the age of 1.45 Ga for EEC.

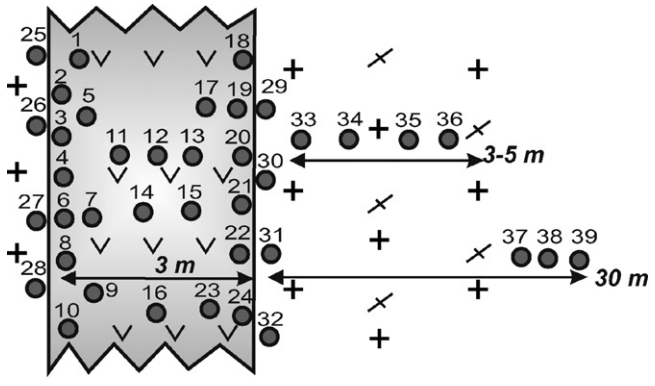


Fig. 11. Sketch map showing the palaeomagnetic sampling for baked contact test of type-A dolerite dyke TM1 from the Tamkhanka island. Samples were taken from the dyke, baked contact zone of the dyke and from varying distances in the unbaked area.

6.4. Comparison of the APWPs for the EEC, Laurentia and Siberia, and Palaeo-Mesoproterozoic supercontinent reconstructions

In order to make a new 1.45 Ga reconstruction for EEC, Laurentia and Siberia, we have compared the APW paths of these continents based on three time slots (Table 3). For EEC we have used the Late Palaeoproterozoic 1780 Ma key pole (VZH, Fig. 17) from the Vazhinka River succession in NW Russia (Pisarevsky and Sokolov, 2001), the 1452 Ma pole (LA) reported here, and the Late Mesoproterozoic 1265 Ma key pole (PJI) from the Postjotnian intrusions in western Finland (see Buchan et al., 2000).

These three key poles for Baltica are compared with similar aged poles from Laurentia (Table 3). The oldest pole CL is from the 1740 ± 5 Ma Cleaver Lake dykes (Irving et al., 2004). For 1.45 Ga we calculated a mean pole, MLR, from the 1460 ± 5 Ma Michikamau Intrusion (Emslie et al., 1976), 1450 ± 5 Harp Lake complex (Buchan et al., 2000) and 1458 ± 49 Ma Tobacco Root Mountains (Harlan et al., 2008) (Table 3, Fig. 17). The high-quality 1476 ± 16 Ma St. Francois Mt. pole (Table 3) of Meert and Stuckey (2002), which is older than the mean MLR pole of Laurentia or the MPR pole of EEC, has been used only for comparison with the Sololi Group pole (SG) for Siberia (Wingate et al., 2009). For the Late Mesoproterozoic, we have used the 1267 ± 2 Ma pole from the Mackenzie dykes (see Buchan et al., 2000) in Laurentia (Table 3, Fig. 17).

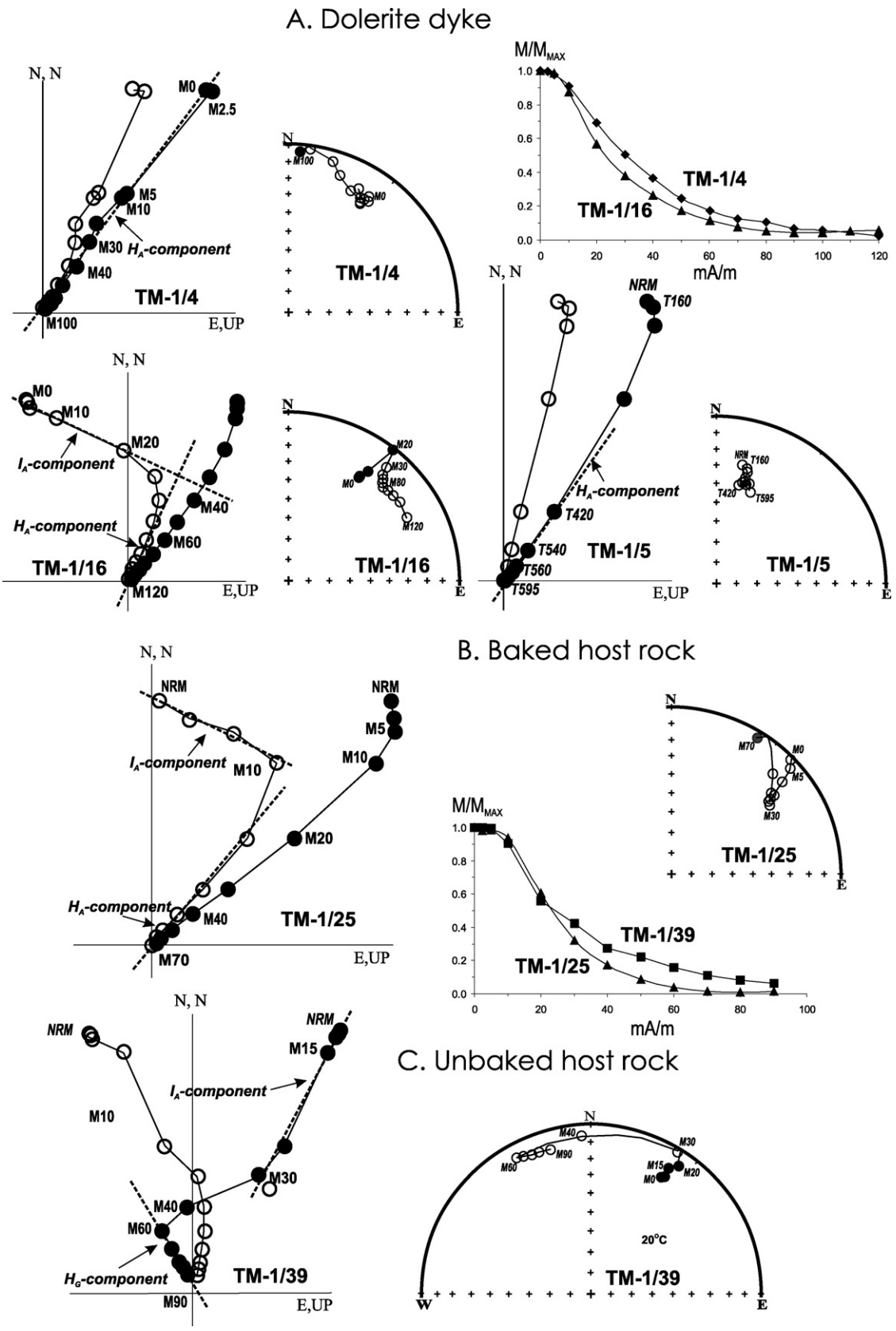


Fig. 12. Examples of AF and thermal demagnetization behaviours of baked contact test for the type-A dolerite dyke TM1. (A) Three examples (TM-1/4, TM-1/16, TM-1/23) from the dolerite dyke, (B) baked host gneiss, (C) unbaked gneiss of the host Ladoga Formation. See location of samples in Fig. 11. Explanations of figures as in Fig. 7.

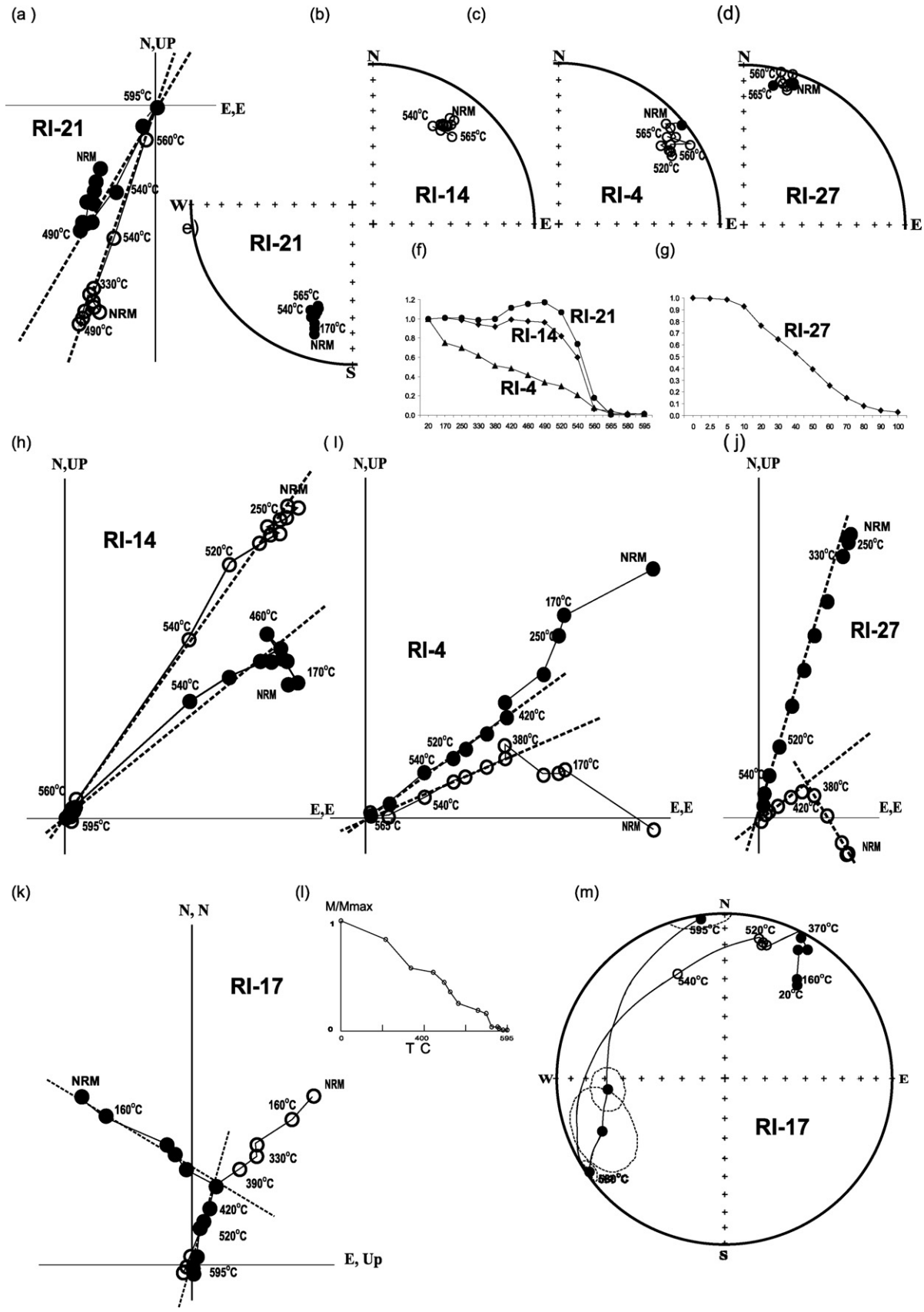


Fig. 13. Examples of thermal and AF demagnetization behaviours of specimens of type-B dolerite dyke from the Riekkalansaari site RA+RI. (a, h–k) orthogonal vector projections, (b–e, m) stereographic projections, (f, g, and l) relative NRM intensity decay curves upon thermal demagnetizations. For explanations, see Fig. 7.

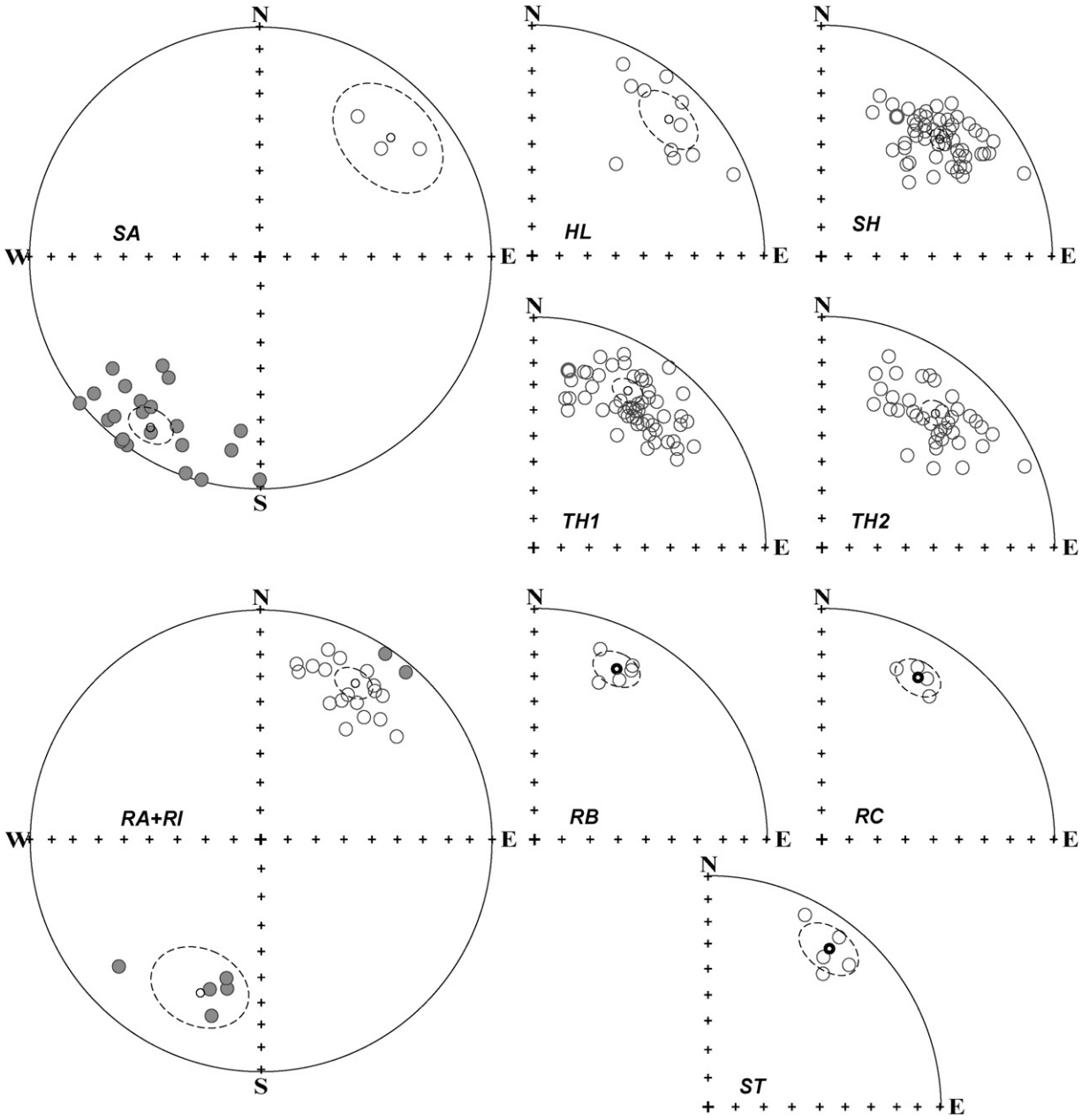


Fig. 14. Sample mean palaeomagnetic directions for the mafic rocks of the Lake Ladoga region. The site mean directions are shown as smaller circle with $\alpha 95$ confidence cones about the means. For abbreviations, see Fig. 2.

When coeval poles from Baltica and Laurentia are paired, a complete matching of the poles is obtained. The reconstruction confirms the invariable Laurentia-Baltica configuration for the period between 1.74 and 1.27 Ga, already proposed by e.g. Buchan et al. (2000) and Salminen and Pesonen (2007). According to the reconstruction, Baltica was located at low southern latitudes with its north-eastern part facing eastern Greenland (Fig. 17). As elaborated by Karlström et al. (2001), but suggested already by Gower et al. (1990) and Gorbatschev and Bogdanova (1993), south-eastern Laurentia and south-western EEC appear to have been sited along one single continuous active continental margin both in the late Palaeoproterozoic and in the Mesoproterozoic.

Wingate et al. (2009) have tested the Laurentia-Siberia connection during middle Mesoproterozoic- to Neoproterozoic, based on the coeval 1476 ± 16 Ma St. Francois Mt. pole and 1473 ± 24 Ma Kyutingde-Sololi (Sololi Group) pole. We have used their Euler rotation parameters for Siberia in relation to Laurentia when reconstructing the Siberia position in Fig. 17. The reconstruction supports the Baltica-Laurentia-Siberia connection (with an open space between the continents) during Mesoproterozoic, as proposed by Wingate et al. (2009) and Pisarevsky et al. (2009). The reconstruction is consistent with geological evidence.

With regard to the continuity of Archaean and Palaeoproterozoic crustal provinces in northern Laurentia and southern Siberia these have been proposed to be connected in the Late Palaeoproterozoic (Condie and Rosen, 1994; Frost et al., 1998; Rainbird et al., 1998; Rosen et al., 2006). According to some Palaeo- to Mesoproterozoic palaeomagnetic reconstructions, southern Siberia is also located fairly close to northern Laurentia (Wu et al., 2005; Veselovskiy, 2006; Didenko et al., 2009) or with a gap that may have been occupied by the still hypothetical Arctida continent (Pisarevsky and Natapov, 2003; Li et al., 2008; Pisarevsky et al., 2008, 2009; Wingate et al., 2009). Furthermore, Wu et al. (2005) have proposed a link between Baltica, Laurentia and Siberia with the North China

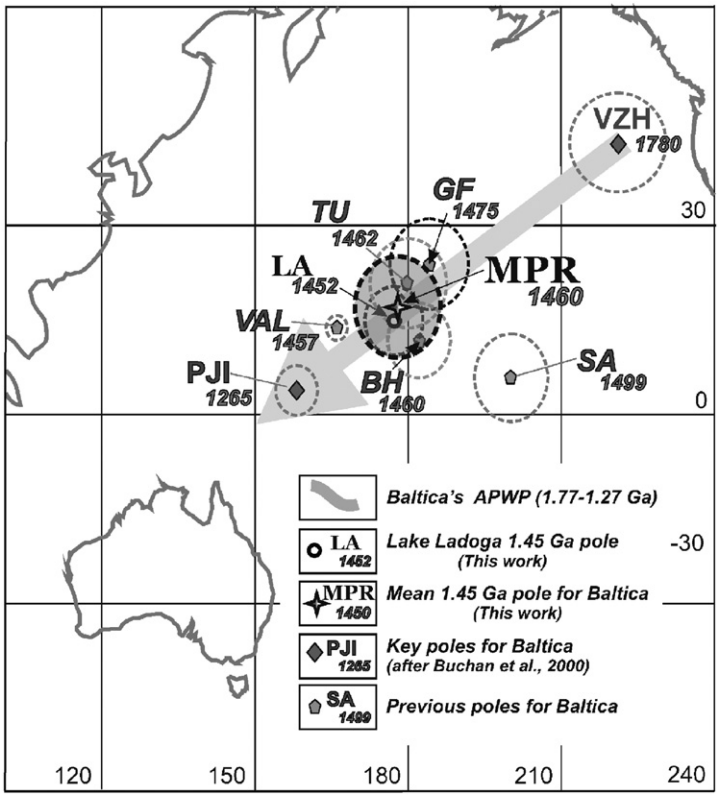


Fig. 15. Palaeomagnetic poles of the high coercivity and high unblocking temperature component H obtained in this study (Table 2) are marked as star and those from previous studies (Table 3) as rhombus. The shaded *A*95% confidence circle indicates the mean pole of this study.

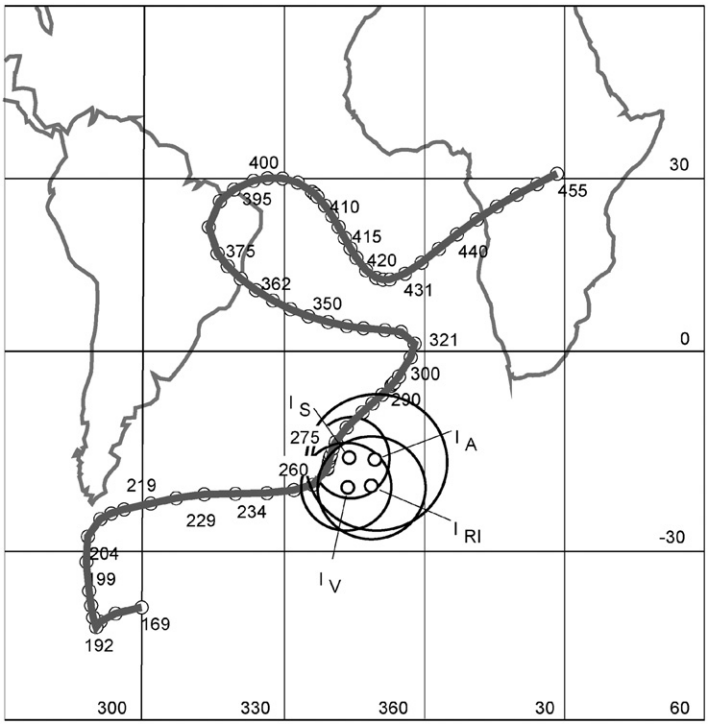


Fig. 16. Palaeomagnetic poles calculated from the direction of the low-intermediate components (Table 2) with *A*95% confidence circles plotted along the Palaeozoic APW path of Baltica (Smethurst et al., 1998). I_S, Salmi basalts, I_V, mean for Valaam sill, I_A, mean for Type-A dykes, I_{RI}, mean for Riekkalansaari dyke (Type-B dyke).

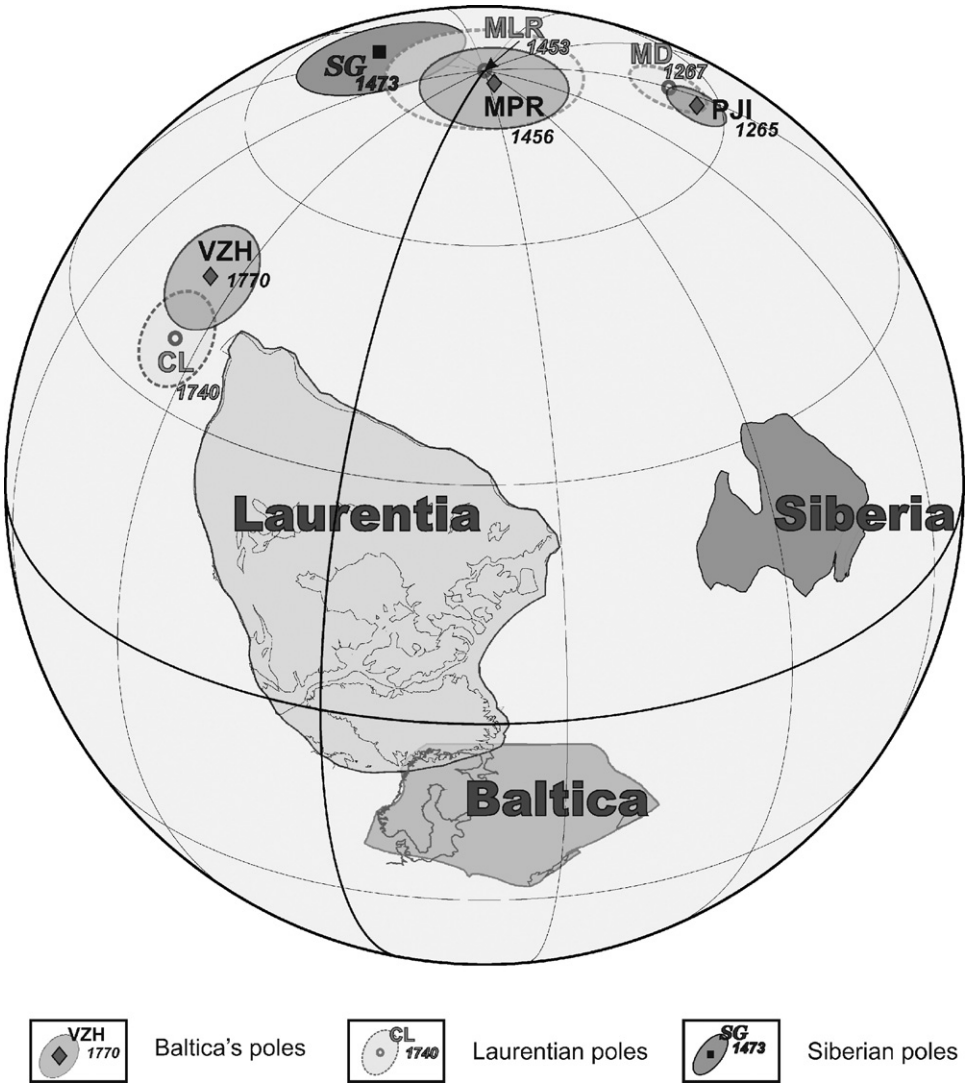


Fig. 17. Reconstruction of the Baltica (EEC), Laurentia and Siberia at ~1452 Ma. Baltica is rotated to Laurentia about an Euler pole (+anticlockwise) at 41.0°N, 357.0°E, +38°; Siberia to Laurentia at 65.0°N 144.0°E +141.8° (Wingate et al., 2009). Laurentia is rotated to the absolute framework about a pole 0°, 123°, +87°, according to the MLR mean pole for ~1450 Ma (Table 3). Numbers near poles indicate their ages in Ma. Abbreviations of palaeopoles are given in Table 3.

craton between 1.80 and 1.35 Ga. Thus, the now available geological, palaeomagnetic and geochronological data for Baltica (EEC), Siberia, Laurentia and possibly North China cratons indicate their shared evolution and a long-lived connection during most of the Mesoproterozoic, when they still formed parts of the supercontinent Columbia.

7. Conclusions

The new baddeleyite U–Pb dating gives an age of 1452 ± 12 Ma for the Fe-rich olivine dolerites at Lake Ladoga, indistinguishable from previously reported U–Pb ages of the Valaam sill. These ages link the mafic rocks in the Lake Ladoga region to a period when local rift basins were developed in Fennoscandia.

Palaeomagnetic data from the magmatic rocks of Lake Ladoga region are of good quality. The principal remanence carries are SD magnetite and low-titanium titanomagnetite. The mafic dykes and the Salmi basalt show both normal and reversed polarities. Positive reversal tests for both formations suggest that these rocks are related to the same magmatic event. The primary origin of the characteristic H-component of the dolerite dykes is supported by a positive baked contact test. Based on this evidence, the paleomagnetic pole can be regarded as a key pole for Baltica at 1452 Ma. The new key pole helps to constrain the position of the EEC (Baltica) in the supercontinent Columbia in the Late Palaeoproteorozoic and Mesoproterozoic, and in general, supports the stability of the EEC-Laurentia-Siberia configuration during that time.

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